

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Exponential Rational Integral

Evaluate the integral:

$$\int_0^{\infty} \frac{(1 - e^x)(1 - e^{-x})}{(1 + e^x)(e^x + e^{-x})(e^{-x} + 1)} dx$$

Solution: First, we can simplify the integrand by algebraically manipulating the exponential terms. Let $y = e^x$. The differentials transform as $dx = \frac{dy}{y}$, and the limits of integration change from $[0, \infty)$ to $[1, \infty)$.

Substitute $e^x = y$ into the integrand:

$$\frac{(1 - y) \left(1 - \frac{1}{y}\right)}{(1 + y) \left(y + \frac{1}{y}\right) \left(\frac{1}{y} + 1\right)} \cdot \frac{1}{y}$$

Simplify the numerator and the denominator by multiplying out the fractions: Numerator:

$$(1 - y) \left(\frac{y - 1}{y}\right) = -\frac{(y - 1)^2}{y}$$

Denominator:

$$(1 + y) \left(\frac{y^2 + 1}{y}\right) \left(\frac{y + 1}{y}\right) = \frac{(y + 1)^2(y^2 + 1)}{y^2}$$

Putting it all together with the $1/y$ from the differential:

$$\frac{-\frac{(y-1)^2}{y}}{\frac{(y+1)^2(y^2+1)}{y^2}} \cdot \frac{1}{y} = -\frac{(y-1)^2}{(y+1)^2(y^2+1)}$$

We now decompose the rational function using partial fractions:

$$-\frac{y^2 - 2y + 1}{(y + 1)^2(y^2 + 1)} = \frac{A}{y + 1} + \frac{B}{(y + 1)^2} + \frac{Cy + D}{y^2 + 1}$$

Solving for the constants, we obtain $A = 0$, $B = -2$, $C = 0$, and $D = 1$. The integral decomposes beautifully into:

$$\int_1^{\infty} \left(\frac{1}{y^2 + 1} - \frac{2}{(y + 1)^2} \right) dy$$

Now, evaluate the antiderivatives:

$$\lim_{R \rightarrow \infty} \left[\arctan(y) + \frac{2}{y + 1} \right]_1^R$$

Evaluate the limits: As $R \rightarrow \infty$, $\arctan(R) \rightarrow \frac{\pi}{2}$ and $\frac{2}{R+1} \rightarrow 0$. At $y = 1$, we get $\arctan(1) + \frac{2}{2} = \frac{\pi}{4} + 1$. Subtracting the lower limit from the upper limit:

$$\frac{\pi}{2} - \left(\frac{\pi}{4} + 1 \right) = \frac{\pi}{4} - 1$$

1.2 Hard Integral

Logarithmic Rational Integral with Geometric Series

Evaluate the integral:

$$\int_0^1 \frac{(1+x) \ln(x)}{1-x^3} dx$$

Solution: We rewrite the integrand by expanding $\frac{1}{1-x^3}$ into an infinite geometric series. Since $x \in (0, 1)$, $|x^3| < 1$, allowing the expansion:

$$\frac{1}{1-x^3} = \sum_{n=0}^{\infty} x^{3n}$$

Multiply this series by the remaining terms in the numerator $(1+x) \ln(x)$:

$$\frac{(1+x) \ln(x)}{1-x^3} = (1+x) \ln(x) \sum_{n=0}^{\infty} x^{3n} = \sum_{n=0}^{\infty} (x^{3n} + x^{3n+1}) \ln(x)$$

Now, interchange the integral and the sum (justified by uniform convergence on $(0, 1)$):

$$\sum_{n=0}^{\infty} \left(\int_0^1 x^{3n} \ln(x) dx + \int_0^1 x^{3n+1} \ln(x) dx \right)$$

Recall the standard identity, which can be derived using integration by parts:

$$\int_0^1 x^k \ln(x) dx = -\frac{1}{(k+1)^2}$$

Applying this identity yields:

$$-\sum_{n=0}^{\infty} \left(\frac{1}{(3n+1)^2} + \frac{1}{(3n+2)^2} \right)$$

Let S be this sum. We know the sum of all inverse squares is given by the Basel problem:

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

Notice that S contains all the terms of the Basel sum *except* those where the index k is a multiple of 3 (i.e., $k = 3n + 3 = 3(n + 1)$). Let's find the sum of the missing terms:

$$\sum_{m=1}^{\infty} \frac{1}{(3m)^2} = \frac{1}{9} \sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{1}{9} \left(\frac{\pi^2}{6} \right) = \frac{\pi^2}{54}$$

Therefore, our required sum S is the total Basel sum minus the multiples of 3:

$$S = \frac{\pi^2}{6} - \frac{\pi^2}{54} = \frac{9\pi^2}{54} - \frac{\pi^2}{54} = \frac{8\pi^2}{54} = \frac{4\pi^2}{27}$$

Including the negative sign from the integration by parts step, the final answer is:

$$-\frac{4\pi^2}{27}$$

Differentials

2.1 Medium Differential

Second-Order Linear ODE

A liquid medicine diffuses through a curved blood vessel. The concentration $C(t)$ satisfies:

$$C''(t) + 9C(t) = \cos(3t)$$

Given the initial conditions $C(0) = 0$ and $C'(0) = 2$, find the concentration at $t = \frac{\pi}{6}$.

Solution: First, we find the general solution to the homogeneous equation $C_h''(t) + 9C_h(t) = 0$. The characteristic equation is $r^2 + 9 = 0$, giving purely imaginary roots $r = \pm 3i$.

$$C_h(t) = A \cos(3t) + B \sin(3t)$$

Next, we determine a particular solution $C_p(t)$. Notice that the forcing function on the right-hand side, $\cos(3t)$, is already a part of the homogeneous solution. This means the system is in resonance, and we must multiply our standard guess by t :

$$C_p(t) = t(D \cos(3t) + E \sin(3t))$$

Taking derivatives:

$$C_p'(t) = D \cos(3t) + E \sin(3t) + t(-3D \sin(3t) + 3E \cos(3t))$$

$$C_p''(t) = -6D \sin(3t) + 6E \cos(3t) - 9t(D \cos(3t) + E \sin(3t))$$

Substitute $C_p(t)$ and $C_p''(t)$ back into the original ODE:

$$-6D \sin(3t) + 6E \cos(3t) - 9C_p(t) + 9C_p(t) = \cos(3t)$$

$$-6D \sin(3t) + 6E \cos(3t) = \cos(3t)$$

Equating coefficients yields $D = 0$ and $6E = 1 \implies E = \frac{1}{6}$. The general solution is:

$$C(t) = A \cos(3t) + B \sin(3t) + \frac{1}{6}t \sin(3t)$$

Now apply the initial conditions. $C(0) = 0 \implies A(1) + 0 + 0 = 0 \implies A = 0$. The derivative is:

$$C'(t) = 3B \cos(3t) + \frac{1}{6} \sin(3t) + \frac{1}{2}t \cos(3t)$$

$C'(0) = 2 \implies 3B(1) + 0 + 0 = 2 \implies B = \frac{2}{3}$. The specific solution is:

$$C(t) = \frac{2}{3} \sin(3t) + \frac{1}{6}t \sin(3t)$$

Finally, evaluate the concentration at $t = \frac{\pi}{6}$:

$$C\left(\frac{\pi}{6}\right) = \frac{2}{3} \sin\left(\frac{\pi}{2}\right) + \frac{1}{6}\left(\frac{\pi}{6}\right) \sin\left(\frac{\pi}{2}\right) = \frac{2}{3}(1) + \frac{\pi}{36}(1) = \frac{2}{3} + \frac{\pi}{36}$$

2.2 Hard Differential

Clairaut's Equation Geometry

Let a curve C in the upper half-plane ($y > 0$) satisfy the differential equation:

$$y^2 + \left(x - \frac{y}{y'}\right)^2 = \left(\int_0^\infty \frac{\ln(1+x^4)}{1+x^2} dx\right)^2$$

Write parametric equations of the singular solution curve C in the form $x = x(t)$, $y = y(t)$, where $t = y'$ represents the slope of the tangent line. Hence, find the exact value of $x(1) + y(1)$.

Solution: Let K be the integral on the right-hand side. Using the substitution $x = \tan \theta$ and known Fourier series expansions for log-trigonometric integrals, it elegantly evaluates to:

$$K = \int_0^\infty \frac{\ln(1+x^4)}{1+x^2} dx = \pi \ln(2 + \sqrt{2})$$

The differential equation can be written as:

$$y^2 + \left(x - \frac{y}{y'}\right)^2 = K^2$$

Geometrically, $x - \frac{y}{y'}$ is the x -intercept of the tangent line to the curve at point (x, y) . The equation states that the distance from the origin $(0, 0)$ to the point (x_{int}, y) is strictly K .

We seek a parametric representation of this singular curve. Let's find a parameterization using an angle θ and test the hypothesis that the slope is exactly $y' = \frac{1}{\theta}$. From the fundamental geometric relations $x - \frac{y}{y'} = K \cos \theta$ and $y = K \sin \theta$, we substitute our theoretical slope $y' = \frac{1}{\theta}$:

$$x - \theta(K \sin \theta) = K \cos \theta \implies x(\theta) = K(\cos \theta + \theta \sin \theta)$$

Let's verify this parameterization perfectly aligns with our initial slope hypothesis by taking the derivative:

$$\frac{dx}{d\theta} = K(-\sin \theta + \sin \theta + \theta \cos \theta) = K\theta \cos \theta$$

$$\frac{dy}{d\theta} = K \cos \theta$$

$$y' = \frac{dy/d\theta}{dx/d\theta} = \frac{K \cos \theta}{K\theta \cos \theta} = \frac{1}{\theta}$$

This flawlessly confirms the mathematical consistency of the system. Therefore, our parameter relates to the slope as $\theta = \frac{1}{t}$. Substituting θ with $\frac{1}{t}$ yields our singular solution parametric equations:

$$x(t) = K \left(\cos \left(\frac{1}{t} \right) + \frac{1}{t} \sin \left(\frac{1}{t} \right) \right)$$

$$y(t) = K \sin \left(\frac{1}{t} \right)$$

We want to evaluate $x(1) + y(1)$. By setting $t = 1$:

$$x(1) = K(\cos(1) + \sin(1))$$

$$y(1) = K \sin(1)$$

Summing them and substituting the value of K :

$$x(1) + y(1) = K(\cos(1) + 2 \sin(1)) = \pi \ln(2 + \sqrt{2})(\cos(1) + 2 \sin(1))$$

Derivatives

3.1 Medium Derivative

Related Rates of a Circumcircle

An isosceles triangle has two equal sides of length $a = 2.4$ m. The angle between these sides θ , increases at a rate of 0.015 rad/s. Find the rate of change of the circumcircle radius R in cm/h when $\theta = \frac{\pi}{3}$.

Solution: Let the sides be a, a , and b , where the angle strictly between the equal sides is θ . By the Law of Cosines, the base b is given by:

$$b = \sqrt{a^2 + a^2 - 2a^2 \cos \theta} = a\sqrt{2(1 - \cos \theta)} = 2a \sin\left(\frac{\theta}{2}\right)$$

The area of the triangle is $K = \frac{1}{2}a^2 \sin \theta$. The circumradius R of any triangle is related to its sides and area by $R = \frac{abc}{4K}$. Substituting our expressions:

$$R = \frac{a \cdot a \cdot b}{4\left(\frac{1}{2}a^2 \sin \theta\right)} = \frac{b}{2 \sin \theta} = \frac{2a \sin(\theta/2)}{2(2 \sin(\theta/2) \cos(\theta/2))} = \frac{a}{2 \cos(\theta/2)} = \frac{a}{2} \sec\left(\frac{\theta}{2}\right)$$

Now, differentiate R with respect to time t using the chain rule:

$$\frac{dR}{dt} = \frac{a}{2} \cdot \sec\left(\frac{\theta}{2}\right) \tan\left(\frac{\theta}{2}\right) \cdot \frac{1}{2} \cdot \frac{d\theta}{dt} = \frac{a}{4} \sec\left(\frac{\theta}{2}\right) \tan\left(\frac{\theta}{2}\right) \frac{d\theta}{dt}$$

We evaluate this at $\theta = \frac{\pi}{3}$, which means $\frac{\theta}{2} = \frac{\pi}{6}$. Given $a = 2.4$ m and $\frac{d\theta}{dt} = 0.015$ rad/s:

$$\sec\left(\frac{\pi}{6}\right) = \frac{2}{\sqrt{3}}, \quad \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$$

Substitute these values into the rate equation:

$$\frac{dR}{dt} = \frac{2.4}{4} \left(\frac{2}{\sqrt{3}}\right) \left(\frac{1}{\sqrt{3}}\right) (0.015) = 0.6 \left(\frac{2}{3}\right) (0.015) = 0.4(0.015) = 0.006 \text{ m/s}$$

Convert this rate into cm/h:

$$\begin{aligned} 0.006 \text{ m/s} &= 0.6 \text{ cm/s} \\ 0.6 \text{ cm/s} \times 3600 \text{ s/h} &= 2160 \text{ cm/h} \end{aligned}$$

3.2 Hard Derivative

Inverse Function Second Derivative

Let $f(x) = \tan\left(\frac{\pi}{4}x\right) + 3e^{\sin(\pi x)} - 3$. Note that $f(0) = 0$. Let $h(x)$ be the inverse function of $f(x)$. Find $1000 \cdot h''(0)$.

Solution: For an inverse function $h(y) = x$ corresponding to $f(x) = y$, the first and second derivatives at a point are given by the standard Inverse Function Theorem formulas:

$$h'(y) = \frac{1}{f'(x)} \quad \text{and} \quad h''(y) = -\frac{f''(x)}{(f'(x))^3}$$

We are given $y = 0$, which clearly maps to $x = 0$ since $f(0) = 0$. First, calculate $f'(x)$:

$$f'(x) = \frac{\pi}{4} \sec^2\left(\frac{\pi}{4}x\right) + 3\pi \cos(\pi x)e^{\sin(\pi x)}$$

Evaluate at $x = 0$:

$$f'(0) = \frac{\pi}{4}(1) + 3\pi(1)(1) = \frac{\pi}{4} + 3\pi = \frac{13\pi}{4}$$

Next, calculate the second derivative $f''(x)$ using product and chain rules:

$$f''(x) = \frac{\pi}{4} \left[2 \sec^2\left(\frac{\pi}{4}x\right) \tan\left(\frac{\pi}{4}x\right) \cdot \frac{\pi}{4} \right] + 3\pi \left[-\pi \sin(\pi x)e^{\sin(\pi x)} + \pi \cos^2(\pi x)e^{\sin(\pi x)} \right]$$

Evaluate at $x = 0$: Since $\tan(0) = 0$ and $\sin(0) = 0$, the initial terms vanish.

$$f''(0) = 0 + 3\pi [0 + \pi(1)^2 e^0] = 3\pi^2$$

Now substitute these back into the inverse second derivative formula:

$$h''(0) = -\frac{3\pi^2}{\left(\frac{13\pi}{4}\right)^3} = -\frac{3\pi^2}{\frac{2197\pi^3}{64}} = -\frac{192}{2197\pi}$$

Finally, multiply the entire outcome by 1000:

$$1000 \cdot h''(0) = -\frac{192000}{2197\pi}$$

Physics & Engineering

4.1 Circuits

RC Circuit Charging Time

A $12.0 \mu\text{F}$ capacitor is charged through a $4.70 \text{ M}\Omega$ resistor from a 24.0 V supply. How long does it take for the capacitor to reach 95.0% of its final charge?

Solution: The time constant τ for an RC circuit is the product of its resistance and capacitance:

$$\tau = RC = (4.70 \times 10^6 \Omega)(12.0 \times 10^{-6} \text{ F}) = 56.4 \text{ s}$$

The charge on the capacitor $Q(t)$ as a function of time is given by the charging equation:

$$Q(t) = Q_{\text{final}}(1 - e^{-t/\tau})$$

We want to find the time t when the charge reaches 95.0% of its final value, so we set $Q(t) = 0.95Q_{\text{final}}$:

$$\begin{aligned} 0.95 &= 1 - e^{-t/\tau} \\ e^{-t/\tau} &= 0.05 \end{aligned}$$

Take the natural logarithm of both sides:

$$-\frac{t}{56.4} = \ln(0.05)$$

$$t = -56.4 \ln(0.05) \approx -56.4(-2.9957) \approx 168.96 \text{ s}$$

Rounding to three significant figures according to the given parameters, it takes **169 s**.

4.2 Mechanics

Critical Spool Rolling Angle

A uniform solid cylinder of mass M and radius R acts as a spool. Wound around its inner axle of radius $r = \frac{1}{2}R$ is a light string. The cylinder is placed on a rough horizontal tabletop where the coefficient of static friction is large enough to prevent slipping. The string emerges from the underside of the inner axle and is pulled with a force F at a constant angle θ to the horizontal. Find the critical angle θ_c (in degrees) at which the linear acceleration of the center of mass is zero for any non-zero F .

Solution: For the spool to have zero linear acceleration ($a = 0$) while rolling without slipping, its angular acceleration must also rigorously be zero ($\alpha = 0$). This condition implies that the net torque about any point is zero. The most strategic point to pick is the instantaneous center of rotation, which is the contact point P between the cylinder and the ground. The normal force, the gravitational force, and the static friction force all inherently pass precisely through the point P . Their respective lever arms are zero, meaning they provide no torque about P . Therefore, for the net torque to be zero, the torque provided strictly by the tension F must also be zero. This fundamentally requires the line of action of the pulling force F to pass exactly through the contact point P . By sketching a simple right triangle linking the center O , the point of tangency T of the string on the inner axle, and the contact point P : The distance

OP is the outer radius R , and the distance OT is the inner radius r . The angle the string makes with the vertical, α , evaluates cleanly:

$$\sin(\alpha) = \frac{r}{R} = \frac{R/2}{R} = \frac{1}{2} \implies \alpha = 30^\circ$$

The angle θ_c that the string makes with the *horizontal* is complementary to this vertical angle. Thus:

$$\theta_c = 90^\circ - 30^\circ = 60^\circ$$

4.3 Quantum

Stopping Potential of Photoelectrons

Monochromatic light of wavelength $\lambda = 300 \text{ nm}$ is incident on a clean metal plate with a work function of $\phi = 2.0 \text{ eV}$. Find the stopping potential required to completely halt the fastest emitted photoelectrons in volts (V). Take $hc = 1240 \text{ eV} \cdot \text{nm}$.

Solution: First, we analytically establish the energy E of the incident monochromatic photons using the Planck-Einstein relation:

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{300 \text{ nm}} = 4.133 \text{ eV}$$

By invoking the conservation of energy inherent in the photoelectric equation, the maximum kinetic energy K_{max} of the ejected photoelectrons strictly equates to the photon energy minus the work function of the metal:

$$K_{max} = E - \phi = 4.133 \text{ eV} - 2.0 \text{ eV} = 2.133 \text{ eV}$$

The stopping potential V_s precisely offsets this maximal kinetic energy, where $K_{max} = eV_s$. Dividing by the elementary charge e , the magnitude of the potential purely matches the kinetic energy expressed in electron-volts:

$$V_s = 2.133 \text{ V}$$

Given the parameter inputs strictly enforce multiple significant figures, the properly rounded answer is **2.13 V**.